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Max Subarray Sum by Removing At Most One

Difficulty: **Medium**Accuracy: **32.53%**Submissions: **35K+**Points: **4**

Given an array `arr[]`, find the maximum sum of a non-empty subarray. You are allowed to skip at most one element in the subarray.

Note: After skipping the element, the subarray must still be non-empty.

Examples:

Input: `arr[] = [1, 2, 3, -4, 5]`

Output: 11

Explanation: We can get maximum sum subarray by skipping -4.

Input: `arr[] = [-2, -3, 4, -1, -2, 1, 5, -3]`

Output: 9

Explanation: We can get maximum sum subarray by skipping -2 as `[4,-1,1,5]` sums to 9, which is the maximum achievable sum.

Constraints:

$1 \leq \text{arr.size()} \leq 10^6$

```
1 class Solution:
2     def maxSumSubarray(self, arr):
3         n = len(arr)
4
5         if n == 1:
6             return arr[0]
7
8         # Forward Kadane
9         fw = [0] * n
10        fw[0] = arr[0]
11
12        for i in range(1, n):
13            fw[i] = max(arr[i], fw[i - 1] + arr[i])
14
15        # Backward Kadane
16        bw = [0] * n
17        bw[n - 1] = arr[n - 1]
18
19        for i in range(n - 2, -1, -1):
20            bw[i] = max(arr[i], bw[i + 1] + arr[i])
21
22        ans = max(fw) # No deletion
23
24        # Try deleting one element
25        for i in range(1, n - 1):
26            ans = max(ans, fw[i - 1] + bw[i + 1])
27
28        return ans
```